dirac matrices: the eight real 16 x 16 Dirac matrices of the author's notebook 4+4.nb

Provenance file of the revision UM-3 (universe_model_3): construction, reality, anticommutation relations, spin(4,4), Spin(4,4) and Pin(4,4), the chiral projection operators, and the proof that every calculation uses these matrices

Matrices and notebook: Patrick L. Nash Provenance: Claude (Anthropic), 2026-10-24

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1 The answer

This file was generated by the program `um3_dirac_matrices_doc.py` of UM-3 from the results of `um3_dirac_matrices.py`. It proves, with exact computations whose results are printed here, the statements

the author asked for on 2026-10-24 about the eight real 16×16 Dirac matrices. It is self-contained: every equation it uses is written here and every symbol is defined right below the equation in which it appears. All 55 named checks of the program `universe_model_3/code/um3_dirac_matrices.py` pass; each statement below names the check that proves it.

1. **They are the matrices of `4+4.nb`.** The eight matrices $T_a =$ `T16^A[a]` ($a = 0, \ldots, 7$) were rebuilt cell by cell from the author's own input cells (quoted verbatim in Section 2) and are equal, entry by entry, to every matrix that the notebook itself stored as an output when Mathematica evaluated those cells: the eight T_a (`Out[210]`), $T_0 T_1 \cdots T_7$ (`Out[98]`), P_L and P_R (`Out[111]`, `Out[112]`) and seven more; 26 stored True/False outputs (283 values) of the matrix cells are reproduced identically. `B11_stored_T16A_0_to_7` (passed), `B8_stored_T16A8` (passed), `B13_stored_boolean_outputs` (passed).
2. **They are the matrices used in this repository's calculations.** The same eight matrices, entry by entry, are used by the author's notebook `Pair_Creation_of_Universes_WaveFunctionOfUniverse-4+4-Einstein-Lovelock-Nash.nb` (the same defining formulas), the Revision record, the universe model UM-2, nine Wolfram packages and the Stage-1 fixture (Section 11); every UM-3 computation takes them from one module, `universe_model_3/code/um3_gammas.py`. `D1_pair_creation_extraction` (passed), `D2_pair_creation_same_formulas` (passed), `D5_repository_wolfram_packages` (passed).
3. **They are real.** Every entry is -1, 0 or $+1$; each T_a is a signed permutation matrix and an orthogonal matrix, symmetric for $a = 0, 1, 2, 3$ and antisymmetric for $a = 4, 5, 6, 7$. `C1_real_integer_entries` (passed), `C2_signed_permutation` (passed), `C4_orthogonal_and_symmetry` (passed).
4. **They satisfy the anticommutation relations** $\{T_a, T_b\} = 2\eta_{ab} I_{16}$ with $\eta = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$: they generate the real Clifford algebra of signature $(4, 4)$, which is the full algebra of real 16×16 matrices. `C3_clifford_relations` (passed), `C30_basis_256` (passed).
5. **Their scaled commutators generate Pin(4,4).** The 28 matrices $S_{ab} = \frac{1}{4}[T_a, T_b]$ ($a < b$) are linearly independent, close under commutation with the structure constants of $so(4, 4)$ and act on the T_c as the vector representation: they are a basis of the Lie algebra $spin(4, 4) \cong so(4, 4)$, which is the Lie algebra of Pin(4,4) (and of Spin(4,4)). Each exponential $\exp(\theta S_{ab})$ is a product of two unit vectors, so it lies in Pin(4,4); the exponentials generate the identity component $\mathrm{Spin}_0(4, 4)$; together with the two reflections T_0 and T_4 they generate all of Pin(4,4), whose four components are identified by exact invariants. Precisely stated: the scaled commutators generate Pin(4,4) as its Lie algebra and, by exponentiation, its identity component; no product of exponentials leaves that component, so the reflections are needed for the other three. `C21_SAB_independent` (passed), `C24_spin44_closure` (passed), `C26_exponentials_two_unit_vectors` (passed), `C28_exponentials_in_identity_component` (passed), `C27_pin44_components` (passed).
6. **The chiral projection operators of the subsection "Left - Handed vs . Right - Handed Fermion Fields (Chirality) ; projection operators" are re-derived and confirmed.** The only matrices that commute with every S_{ab} are the combinations of I_{16} and $\Gamma = T_0 T_1 \cdots T_7$; $\Gamma^2 = I_{16}$; the only non-trivial projectors in that algebra are $\frac{1}{2}(I_{16} \mp \Gamma)$, which are the author's P_L and P_R, entry by entry; $\Gamma = \mathrm{diag}(-I_8, I_8)$, so P_L projects on $\Psi_0, \ldots, \Psi_7$ and P_R on $\Psi_8, \ldots, \Psi_{15}$. `C17_commutant_spin44` (passed), `C20_idempotents_of_chirality_algebra` (passed), `B9_stored_PL` (passed), `B10_stored_PR` (passed).
7. **The coordinates are the author's** $\{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$: the input cell `In[48]` of the Section DEFCHART of `Dirac_Einstein-Lovelock_4+4_Nash.nb` is `X={x0[], x1[], x2[], x3[], x4[], x5[], x6[], x7[]}` (spaces added after the commas), and `4+4.nb` uses the same list (its cell 9). The frame index a of T_a is the index of the coordinate x_a. `A6_coordinates_DEFCHART` (passed), `A5_coordinates_in_4p4` (passed).

Each of the eight matrices, the chirality matrix, the Dirac-adjoint matrix, the projectors, the complex structure and the 28 generators is displayed in full below.

2 Notation

The frame metric of the eight-dimensional space is

$$\eta_{ab} = \mathrm{diag}(+1,\ +1,\ +1,\ +1,\ -1,\ -1,\ -1,\ -1)_{ab}, \qquad a, b = 0, 1, \ldots, 7$$

Symbols in this equation.

- η_{ab}: The flat frame metric, $\eta = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ in the order $x_0,\ldots,x_7$ (the author's `eta4488`); η^{ab} is the same matrix.
- a, b, c, d: Frame (orthonormal) indices running over $0, 1, \ldots, 7$; the value k belongs to the author's coordinate x_k (x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the extra time-like directions). No sum over a repeated frame index unless a sum is written.

Four directions are space-like (x_0, x_1, x_2, x_3) and four are time-like (x_4, x_5, x_6, x_7), in the order of the author's coordinates:

$$X = \{x_0,\ x_1,\ x_2,\ x_3,\ x_4,\ x_5,\ x_6,\ x_7\}$$

Symbols in this equation.

- $x_0,\ldots,x_7$: The author's coordinates $X = \{x_0, x_1, \ldots, x_7\}$ (notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb`, Section DEFCHART, `X = {x0[], x1[], ..., x7[]}`).

Matrices act on columns $\Psi = (\Psi_0,\ldots,\Psi_{15})^{\mathsf{T}}$; rows and columns of a 16×16 matrix are counted from 0 in the text and in the displays (the notebook counts them from 1). A block matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ is the author's `ArrayFlatten[{{A, B}, {C, D}}]`.

3 The source: the author's notebook 4+4.nb

The notebook `universe_model_3/author_inputs/4+4.nb` ("eight, real-valued, 16 by 16, Dirac Matrices", Wolfram 15.0) has the sha256 `7ec1709a0028bce0b443f116cf27b5d3e670de7aa195659a9c3274527270429d` and 806 cells; it is read without Mathematica by `universe_model_3/tools/nb_reader.py`, which parses the notebook file (a Wolfram Language expression) and converts every cell to linear text. `A1_notebook_sha256` (passed), `A2_notebook_cell_count` (passed).

The matrices are defined by the following input cells, quoted verbatim (cell number in the file, label, text; `A3_defining_cells_verbatim` (passed)):

```
cell 10  In[7]:=
    ID4=IdentityMatrix[4];
    ID8=IdentityMatrix[8];
cell 11  In[9]:=
    (\[Eta]4488 = ArrayFlatten[{{IdentityMatrix[4], 0}, {0, -IdentityMatrix[4]}}]) // MatrixForm
cell 16  In[18]:=
    \[Sigma]=\[Sigma]88=ArrayFlatten[{{ConstantArray[0,{4,4}],IdentityMatrix[4]},{IdentityMatrix[4],Cons⌋
    ↪ tantArray[0,{4,4}]}}];Protect[\[Sigma]88];
cell 17  In[19]:=
    Zero16=ConstantArray[0,{16,16}];
    G16=ArrayFlatten[{{\[Eta]8,0},{0,-\[Eta]8}}];
    ID16=Id16=IdentityMatrix[16];MId16=ExpandAll[-1*Id16];
    Simp16={Zero16->ZERO16,Id16->ONE16,MId16->MONE16};
cell 24  In[29]:=
    Qa[h_,p_,q_]:=Signature[{h,p,q,4}];
    Qb[h_,p_,q_]:= ID4[[p,4]]*ID4[[q,h]]-ID4[[p,h]]*ID4[[q,4]];
    SelfDualAntiSymmetric[h_,p_,q_]:=Qa[h,p,q] -Qb[h,p,q] ;
    AntiSelfDualAntiSymmetric[h_,p_,q_]:= (Qa[h,p,q] + Qb[h,p,q] );
cell 29  In[35]:=
    Do[s4by4[h]=Table[Table[SelfDualAntiSymmetric[h,p,q], {q, 4}],{p, 4}], {h,1, 3} ]
cell 30  In[36]:=
    Do[t4by4[h]=Table[Table[AntiSelfDualAntiSymmetric[h,p,q], {q, 4}],{p, 4}], {h,1, 3} ]
cell 73  In[60]:=
    \[Zeta][0]=ID8;
    Table[\[Zeta][7-h]= ArrayFlatten[{{0,t4by4[h]},{-t4by4[h],0}}], {h,1, 3} ];
    Table[\[Zeta][h]= ArrayFlatten[{{0,s4by4[h]},{s4by4[h],0}}], {h,1, 3} ];
    (\[Zeta][7]=\[Zeta][1].\[Zeta][2].\[Zeta][3].\[Zeta][4].\[Zeta][5].\[Zeta][6])//MatrixForm
cell 87  In[70]:=
```

```
    \[Tau][0]=ID8;
    Do[{\[Tau][A] = FullSimplify[ExpandAll[(\[Zeta][A])]]} , {A, 1, 7}]
    (*Do[{\[Tau][A] = FullSimplify[ExpandAll[-(\[Sigma].\[Zeta][1].\[Zeta][2].\[Zeta][3].\[Zeta][A])]]} ,
    ↪  {A, 1, 7}]*)
cell 91  In[74]:=
    Overscript[\[Tau], _][0] =ID8;
    Do[{Overscript[\[Tau], _][A] = FullSimplify[ExpandAll[(\[Sigma].Transpose[\[Tau][A]].\[Sigma])]]} ,
    ↪  {A, 1, 7}]
cell 131  In[96]:=
    (\[Sigma]16=T16^A[0].T16^A[1].T16^A[2].T16^A[3])//MatrixForm
cell 133  In[97]:=
    Table[T16^A[A1]= ArrayFlatten[{{0,Overscript[\[Tau], _][A1]},{\[Tau][A1],0}}], {A1,0, 7} ];
cell 134  In[98]:=
    (T16^A[8]=T16^A[0].T16^A[1].T16^A[2].T16^A[3].T16^A[4].T16^A[5].T16^A[6].T16^A[7])//MatrixForm
cell 161  In[111]:=
    (Subscript[P, L]=1/2(ID16-T16^A[8]))//MatrixForm
cell 163  In[112]:=
    (Subscript[P, R]=1/2(ID16+T16^A[8]))//MatrixForm
```

The cells 160 to 166 form the subsection "Left - Handed vs . Right - Handed Fermion Fields (Chirality) ; projection operators" (`A4_chirality_subsection_found` (passed)): cell 161 defines P_L, cell 163 defines P_R and cell 165 tests them; Section 9 below re-derives them.

In cell 131 (`In[96]`) the author assigns `sigma16 = T16^A[0].T16^A[1].T16^A[2].T16^A[3]` before the matrices are defined in cell 133 (`In[97]`): the symbol then holds the unevaluated product, which takes its value as soon as the `T16^A` exist; cell 148 (`In[105]`) confirms the value $\mathrm{diag}(-\sigma, \sigma)$. In cell 87 (`In[70]`) the line inside `(* ... *)` is a comment and is not evaluated.

4 The construction, step by step

4.1 The two 8×8 metrics

$$\eta = \begin{pmatrix} I_4 & 0 \\ 0 & -I_4 \end{pmatrix} = \begin{pmatrix} 1&0&0&0&0&0&0&0 \\ 0&1&0&0&0&0&0&0 \\ 0&0&1&0&0&0&0&0 \\ 0&0&0&1&0&0&0&0 \\ 0&0&0&0&-1&0&0&0 \\ 0&0&0&0&0&-1&0&0 \\ 0&0&0&0&0&0&-1&0 \\ 0&0&0&0&0&0&0&-1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix} = \begin{pmatrix} 0&0&0&0&1&0&0&0 \\ 0&0&0&0&0&1&0&0 \\ 0&0&0&0&0&0&1&0 \\ 0&0&0&0&0&0&0&1 \\ 1&0&0&0&0&0&0&0 \\ 0&1&0&0&0&0&0&0 \\ 0&0&1&0&0&0&0&0 \\ 0&0&0&1&0&0&0&0 \end{pmatrix}$$

Symbols in this equation.

- η_{ab}: The flat frame metric, $\eta = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ in the order $x_0, \ldots, x_7$ (the author's `eta4488`); η^{ab} is the same matrix.
- σ: The real symmetric 8×8 matrix $\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}$ (the author's `sigma88`), with $\sigma^2 = I_8$.
- I_n: The $n \times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).
- 0: The zero matrix of the size required by its position (in a block matrix: a zero block).

(`B1_stored_eta4488` (passed).) The matrix σ has the eigenvalues $+1$ and -1, four times each, so it is also a metric of signature $(4, 4)$.

4.2 The self-dual and anti-self-dual 4×4 matrices

The author's cell 24 defines, for $h = 1, 2, 3$ and $p, q = 1, \ldots, 4$,

$$(s_h)_{pq} = \varepsilon_{hpq4} - \big(\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}\big), \qquad (t_h)_{pq} = \varepsilon_{hpq4} + \big(\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}\big)$$

Symbols in this equation.

- $s_h,\ t_h$: The author's 4×4 matrices `s4by4[h]` and `t4by4[h]` ($h=1,2,3$): real, antisymmetric, self-dual (s_h) and anti-self-dual (t_h) in the four-dimensional sense.
- h,p,q: Integer labels of the notebook: $h=1,2,3$ numbers the matrices s_h, t_h; $p,q=1,\ldots,4$ are their row and column (counted from 1 as in Mathematica).
- ε_{hpqr}: The Levi-Civita symbol of four indices $1,\ldots,4$ (the author's `Signature[{h,p,q,r}]`): $+1$ for an even permutation of $(1,2,3,4)$, -1 for an odd one, 0 if an index repeats.
- δ_{pq}: The Kronecker delta: 1 if $p=q$, else 0 (the author's `ID4[[p,q]]`).

which gives (`B2_stored_s4by4_t4by4` (passed))

$$s_1=\begin{pmatrix}0&0&0&1\\0&0&1&0\\0&-1&0&0\\-1&0&0&0\end{pmatrix},\quad s_2=\begin{pmatrix}0&0&-1&0\\0&0&0&1\\1&0&0&0\\0&-1&0&0\end{pmatrix},\quad s_3=\begin{pmatrix}0&1&0&0\\-1&0&0&0\\0&0&0&1\\0&0&-1&0\end{pmatrix}$$

$$t_1=\begin{pmatrix}0&0&0&-1\\0&0&1&0\\0&-1&0&0\\1&0&0&0\end{pmatrix},\quad t_2=\begin{pmatrix}0&0&-1&0\\0&0&0&-1\\1&0&0&0\\0&1&0&0\end{pmatrix},\quad t_3=\begin{pmatrix}0&1&0&0\\-1&0&0&0\\0&0&0&-1\\0&0&1&0\end{pmatrix}$$

Symbols in this equation.

- $s_h,\ t_h$: The author's 4×4 matrices `s4by4[h]` and `t4by4[h]` ($h=1,2,3$): real, antisymmetric, self-dual (s_h) and anti-self-dual (t_h) in the four-dimensional sense.

Each s_h equals its four-dimensional dual, $\frac{1}{2}\sum_{j_1,j_2}\varepsilon_{pqj_1j_2}(s_h)_{j_1j_2}=(s_h)_{pq}$, and each t_h minus its dual (the author's cells 67 and 70, both giving zero matrices).

4.3 The 8×8 matrices τ_a and $\bar{\tau}_a$

The author's cells 73 and 87 define

$$\tau_0=I_8,\qquad \tau_h=\begin{pmatrix}0&s_h\\s_h&0\end{pmatrix},\qquad \tau_{7-h}=\begin{pmatrix}0&t_h\\-t_h&0\end{pmatrix}\qquad (h=1,2,3),$$

$$\tau_7=\tau_1\tau_2\tau_3\tau_4\tau_5\tau_6$$

Symbols in this equation.

- τ_a: The author's real 8×8 matrices `tau[a]` ($a=0,\ldots,7$), $\tau_a=\zeta_a$.
- ζ_A: The author's intermediate 8×8 matrices `zeta[A]` ($A=0,\ldots,7$), built from s_h and t_h.
- $s_h,\ t_h$: The author's 4×4 matrices `s4by4[h]` and `t4by4[h]` ($h=1,2,3$): real, antisymmetric, self-dual (s_h) and anti-self-dual (t_h) in the four-dimensional sense.
- I_n: The $n\times n$ identity matrix ($n=4,8,16$; the author's `ID4`, `ID8`, `ID16`).
- 0: The zero matrix of the size required by its position (in a block matrix: a zero block).

(the author first names them ζ_A and then sets $\tau_A=\zeta_A$), and cell 91 defines

$$\bar{\tau}_0=I_8,\qquad \bar{\tau}_a=\sigma\,\tau_a^{\mathsf{T}}\,\sigma\quad (a=1,\ldots,7)$$

Symbols in this equation.

- $\bar{\tau}_a$: The author's `taubar[a]` (written $\overline{\tau}$ in the notebook), $\bar{\tau}_0=I_8$, $\bar{\tau}_a=\sigma\,\tau_a^{\mathsf{T}}\,\sigma$ for $a=1,\ldots,7$ (not a complex conjugate).
- τ_a: The author's real 8×8 matrices `tau[a]` ($a=0,\ldots,7$), $\tau_a=\zeta_a$.

- σ: The real symmetric 8×8 matrix $\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}$ (the author's `sigma88`), with $\sigma^2 = I_8$.
- X^{T}, $X^{\dagger}$: The transpose of a matrix and its Hermitian conjugate (transpose of the complex conjugate); for the real matrices of this document they coincide.
- I_n: The $n \times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).

(`B4_stored_tau7` (passed), `C6_taubar_definition` (passed)). Explicitly:

$$\tau_0 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \bar{\tau}_0 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\tau_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \bar{\tau}_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\tau_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \bar{\tau}_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\tau_3 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \bar{\tau}_3 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\tau_4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \bar{\tau}_4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\tau_5 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \bar{\tau}_5 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\tau_6 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \bar{\tau}_6 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\tau_7 = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \bar{\tau}_7 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}$$

Symbols in this equation.

- τ_a: The author's real 8×8 matrices `tau[a]` ($a = 0, \ldots, 7$), $\tau_a = \zeta_a$.
- $\bar{\tau}_a$: The author's `taubar[a]` (written $\overline{\tau}$ in the notebook), $\bar{\tau}_0 = I_8$, $\bar{\tau}_a = \sigma\, \tau_a^{\mathsf{T}}\, \sigma$ for $a = 1, \ldots, 7$ (not a complex conjugate).

They satisfy, for all $a, b = 0, \ldots, 7$ (`C5_block_relations` (passed)),

$$\tau_a \bar{\tau}_b + \tau_b \bar{\tau}_a = 2\eta_{ab} I_8, \qquad \bar{\tau}_a \tau_b + \bar{\tau}_b \tau_a = 2\eta_{ab} I_8$$

Symbols in this equation.

- τ_a: The author's real 8×8 matrices `tau[a]` ($a = 0, \ldots, 7$), $\tau_a = \zeta_a$.
- $\bar{\tau}_a$: The author's `taubar[a]` (written $\overline{\tau}$ in the notebook), $\bar{\tau}_0 = I_8$, $\bar{\tau}_a = \sigma\, \tau_a^{\mathsf{T}}\, \sigma$ for $a = 1, \ldots, 7$ (not a complex conjugate).
- η_{ab}: The flat frame metric, $\eta = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ in the order $x_0, \ldots, x_7$ (the author's `eta4488`); η^{ab} is the same matrix.
- I_n: The $n \times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).

and each τ_a is orthogonal, $\tau_a \tau_a^{\mathsf{T}} = I_8$ (`C7_tau_orthogonal` (passed)).

4.4 The eight 16×16 Dirac matrices

The author's cell 133 (`In[97]`) defines, for $a = 0, \ldots, 7$,

$$T_a = \begin{pmatrix} 0 & \bar{\tau}_a \\ \tau_a & 0 \end{pmatrix}$$

Symbols in this equation.

- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- τ_a: The author's real 8×8 matrices `tau[a]` ($a = 0, \ldots, 7$), $\tau_a = \zeta_a$.
- $\bar{\tau}_a$: The author's `taubar[a]` (written $\overline{\tau}$ in the notebook), $\bar{\tau}_0 = I_8$, $\bar{\tau}_a = \sigma\, \tau_a^{\mathsf{T}}\, \sigma$ for $a = 1, \ldots, 7$ (not a complex conjugate).
- 0: The zero matrix of the size required by its position (in a block matrix: a zero block).

Here they are, every entry (`B11_stored_T16A_0_to_7` (passed)). Below each matrix, its action as a signed permutation is listed in the form "row:sign column" (the only non-zero entry of each row; for example 0:+8 means that row 0 has the entry +1 in column 8).

T_0 = `T16^A[0]`, the Dirac matrix of the direction x_0 (the hidden space-like direction); $T_0^2 = I_{16}$.

$$T_0 = \begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

Signed permutation of T_0: 0:+8, 1:+9, 2:+10, 3:+11, 4:+12, 5:+13, 6:+14, 7:+15, 8:+0, 9:+1, 10:+2, 11:+3, 12:+4, 13:+5, 14:+6, 15:+7.

T_1 = `T16^A[1]`, the Dirac matrix of the direction x_1 (ordinary space); $T_1^2 = I_{16}$.

$$T_1 = \begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

Signed permutation of T_1: 0:-15, 1:-14, 2:+13, 3:+12, 4:-11, 5:-10, 6:+9, 7:+8, 8:+7, 9:+6, 10:-5, 11:-4, 12:+3, 13:+2, 14:-1, 15:-0.

T_2 = `T16^A[2]`, the Dirac matrix of the direction x_2 (ordinary space); $T_2^2 = I_{16}$.

$$T_2 = \begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

Signed permutation of T_2: 0:+14, 1:-15, 2:-12, 3:+13, 4:+10, 5:-11, 6:-8, 7:+9, 8:-6, 9:+7, 10:+4, 11:-5, 12:-2, 13:+3, 14:+0, 15:-1.

T_3 = `T16^A[3]`, the Dirac matrix of the direction x_3 (ordinary space); $T_3^2 = I_{16}$.

$$T_3 = \begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

Signed permutation of T_3: 0:-13, 1:+12, 2:-15, 3:+14, 4:-9, 5:+8, 6:-11, 7:+10, 8:+5, 9:-4, 10:+7, 11:-6, 12:+1, 13:-0, 14:+3, 15:-2.

T_4 = `T16^A[4]`, the Dirac matrix of the direction x_4 (the time); $T_4^2 = -I_{16}$.

$$T_4 = \begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

Signed permutation of T_4: 0:-13, 1:+12, 2:+15, 3:-14, 4:+9, 5:-8, 6:-11, 7:+10, 8:+5, 9:-4, 10:-7, 11:+6, 12:-1, 13:+0, 14:+3, 15:-2.

T_5 = `T16^A[5]`, the Dirac matrix of the direction x_5 (an extra time-like direction); $T_5^2 = -I_{16}$.

$$T_5 = \begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

Signed permutation of T_5: 0:+14, 1:+15, 2:-12, 3:-13, 4:-10, 5:-11, 6:+8, 7:+9, 8:-6, 9:-7, 10:+4, 11:+5, 12:+2, 13:+3, 14:-0, 15:-1.

T_6 = `T16^A[6]`, the Dirac matrix of the direction x_6 (an extra time-like direction); $T_6^2 = -I_{16}$.

$$T_6 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Signed permutation of T_6: 0:+15, 1:-14, 2:+13, 3:-12, 4:-11, 5:+10, 6:-9, 7:+8, 8:-7, 9:+6, 10:-5, 11:+4, 12:+3, 13:-2, 14:+1, 15:-0.

T_7 = `T16^A[7]`, the Dirac matrix of the direction x_7 (an extra time-like direction); $T_7^2 = -I_{16}$.

$$T_7 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Signed permutation of T_7: 0:+8, 1:+9, 2:+10, 3:+11, 4:-12, 5:-13, 6:-14, 7:-15, 8:-0, 9:-1, 10:-2, 11:-3, 12:+4, 13:+5, 14:+6, 15:+7.

Symbols in this equation.

- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- I_n: The $n \times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).

5 Proof that the eight matrices are real

Every step of the construction (integer entries $0, \pm 1$ of ε and δ, block assembly, transposition, products of integer matrices, multiplication by σ) maps integer matrices to integer matrices, so the T_a are real (indeed integer) matrices. The computed entries are only -1, 0 and $+1$ (`C1_real_integer_entries` (passed)); each row and each column of

each T_a holds exactly one non-zero entry, so every T_a is a signed permutation matrix (`C2_signed_permutation` (passed)), hence orthogonal, and (`C4_orthogonal_and_symmetry` (passed))

$$T_a^{\mathsf{T}} = \eta_{aa}\, T_a, \qquad T_a T_a^{\mathsf{T}} = I_{16} \qquad (a = 0, \ldots, 7)$$

Symbols in this equation.

- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- X^{T}, $X^{\dagger}$: The transpose of a matrix and its Hermitian conjugate (transpose of the complex conjugate); for the real matrices of this document they coincide.
- η_{ab}: The flat frame metric, $\eta = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ in the order $x_0, \ldots, x_7$ (the author's `eta4488`); η^{ab} is the same matrix.
- I_n: The $n \times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).

so T_0, T_1, T_2, T_3 are symmetric and T_4, T_5, T_6, T_7 antisymmetric (the author's `Out[102]`, `Out[103]`). Because the matrices are real, $T_a^{\dagger} = T_a^{\mathsf{T}}$.

6 Proof of the anticommutation relations

Theorem. For all $a, b = 0, \ldots, 7$

$$\{T_a, T_b\} = T_a T_b + T_b T_a = 2\, \eta_{ab}\, I_{16}$$

Symbols in this equation.

- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- $[X, Y]$, $\{X, Y\}$: The commutator $XY - YX$ and the anticommutator $XY + YX$ of two matrices.
- η_{ab}: The flat frame metric, $\eta = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ in the order $x_0, \ldots, x_7$ (the author's `eta4488`); η^{ab} is the same matrix.
- I_n: The $n \times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).

Proof. With the block form of T_a,

$$T_a T_b + T_b T_a = \begin{pmatrix} \bar{\tau}_a \tau_b + \bar{\tau}_b \tau_a & 0 \\ 0 & \tau_a \bar{\tau}_b + \tau_b \bar{\tau}_a \end{pmatrix}$$

Symbols in this equation.

- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- τ_a: The author's real 8×8 matrices `tau[a]` ($a = 0, \ldots, 7$), $\tau_a = \zeta_a$.
- $\bar{\tau}_a$: The author's `taubar[a]` (written $\overline{\tau}$ in the notebook), $\bar{\tau}_0 = I_8$, $\bar{\tau}_a = \sigma\, \tau_a^{\mathsf{T}}\, \sigma$ for $a = 1, \ldots, 7$ (not a complex conjugate).
- 0: The zero matrix of the size required by its position (in a block matrix: a zero block).

and the two diagonal blocks are $2\eta_{ab} I_8$ by the 8×8 relations of the previous section (`C5_block_relations` (passed)). Those relations are finitely many identities between explicit integer matrices; they were verified by exact matrix multiplication for every one of the 64 pairs (a, b), and so were the 64 relations of the theorem directly (`C3_clifford_relations` (passed)). An exhaustive exact verification of a finite set of identities between explicit integer matrices is a complete proof. The author's cell 156 (`In[109]`) states the same relations and its stored output is True in all 64 places (`B13_stored_boolean_outputs` (passed)). □

In particular $T_a^2 = +I_{16}$ for the space-like directions x_0, x_1, x_2, x_3 and $T_a^2 = -I_{16}$ for the time-like directions x_4, x_5, x_6, x_7, and different T_a anticommute. The $2^8 = 256$ ordered products $T_{a_1} \cdots T_{a_k}$ ($a_1 < \cdots < a_k$, the empty product being I_{16}) are linearly independent (`C30_basis_256` (passed)): the real algebra generated by the T_a is

the whole algebra of real 16×16 matrices, the Clifford algebra $Cl(4,4) \cong M_{16}(\mathbb{R})$; of the 256 products 136 are symmetric and 120 antisymmetric, as the author found (`Out[205]`, `Out[217]`, `Out[220]`).

7 The Dirac-adjoint matrix σ_{16}

The author's `sigma16` is

$$\sigma_{16} = T_0T_1T_2T_3 = \begin{pmatrix} -\sigma & 0 \\ 0 & \sigma \end{pmatrix} = \begin{pmatrix}
0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0
\end{pmatrix}$$

Symbols in this equation.

- σ_{16}: The author's `sigma16` $= T_0T_1T_2T_3$, the matrix of the Dirac adjoint $\bar{\Psi} = \Psi^\dagger\sigma_{16}$.
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- σ: The real symmetric 8×8 matrix $\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}$ (the author's `sigma88`), with $\sigma^2 = I_8$.
- 0: The zero matrix of the size required by its position (in a block matrix: a zero block).

It is real, symmetric and its own inverse, and (`C8_sigma16` (passed), `C9_sigma16_conjugation` (passed))

$$(\sigma_{16}T_a)^{\mathsf{T}} = -\sigma_{16}T_a, \qquad \sigma_{16}\,T_a\,\sigma_{16}^{-1} = -T_a^{\mathsf{T}} = -T_a^\dagger \qquad (a = 0, \ldots, 7)$$

Symbols in this equation.

- σ_{16}: The author's `sigma16` $= T_0T_1T_2T_3$, the matrix of the Dirac adjoint $\bar{\Psi} = \Psi^\dagger\sigma_{16}$.
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- X^{T}, $X^\dagger$: The transpose of a matrix and its Hermitian conjugate (transpose of the complex conjugate); for the real matrices of this document they coincide.

which is the author's check `In[104]` (all True). The second relation is what makes $\bar{\Psi} = \Psi^\dagger\sigma_{16}$ the Dirac adjoint: under $\Psi \to g\Psi$ with $g = \exp(\frac{1}{2}\omega^{ab}S_{ab})$ the bilinear $\bar{\Psi}\Psi$ is invariant and $\bar{\Psi}T_a\Psi$ transforms as a vector (because $\sigma_{16}S_{ab}\sigma_{16}^{-1} = -S_{ab}^\dagger$ follows from it).

$$\bar{\Psi} = \Psi^\dagger\,\sigma_{16}$$

Symbols in this equation.

- $\bar{\Psi}$: The Dirac adjoint $\bar{\Psi} = \Psi^\dagger\sigma_{16}$.
- Ψ: The 16-component dirac16complex spinor field (components $\Psi_0, \ldots, \Psi_{15}$), on which the matrices act.

- σ_{16}: The author's `sigma16` $= T_0T_1T_2T_3$, the matrix of the Dirac adjoint $\bar\Psi = \Psi^\dagger\sigma_{16}$.
- $X^{\mathsf T}$, $X^\dagger$: The transpose of a matrix and its Hermitian conjugate (transpose of the complex conjugate); for the real matrices of this document they coincide.

8 The scaled commutators: spin(4,4), Spin(4,4) and Pin(4,4)

8.1 Definitions

A **vector** of the Clifford algebra is $v = v^aT_a$ with real v^a; then $v^2 = \eta_{ab}v^av^b\,I_{16}$, and v is a **unit vector** if $v^2 = \pm I_{16}$.

$$v = \sum_{a=0}^{7} v^aT_a, \qquad v^2 = \Big(\sum_{a,b}\eta_{ab}v^av^b\Big)I_{16}$$

Symbols in this equation.

- v, v^a: A vector of the Clifford algebra, $v = v^aT_a$ with real components v^a.
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0,\dots,7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- η_{ab}: The flat frame metric, $\eta = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ in the order $x_0,\dots,x_7$ (the author's `eta4488`); η^{ab} is the same matrix.
- I_n: The $n\times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).

Pin(4,4) is the group generated by the unit vectors (products $u_1u_2\cdots u_k$ of unit vectors); **Spin(4,4)** is its subgroup of products of an even number of unit vectors. The **twisted adjoint action**

$$\rho(g)\,v = \alpha(g)\,v\,g^{-1}$$

Symbols in this equation.

- $\rho(g)$: The twisted adjoint action of g on vectors, $\rho(g)v = \alpha(g)\,v\,g^{-1}$, an element of $O(4,4)$.
- g: An element of the group $\mathrm{Pin}(4,4)$ (an invertible 16×16 real matrix).
- $\alpha(g)$: The parity automorphism: $\alpha(g) = g$ for an even element (a product of an even number of vectors), $\alpha(g) = -g$ for an odd one.
- v, v^a: A vector of the Clifford algebra, $v = v^aT_a$ with real components v^a.

maps vectors to vectors and preserves v^2, so $\rho(g)\in O(4,4)$; for a unit vector u it is the reflection in the hyperplane orthogonal to u, and ρ is a two-to-one homomorphism of Pin(4,4) onto $O(4,4)$ (a standard theorem: Cartan-Dieudonne together with $\ker\rho = \{\pm I_{16}\}$). The Lie algebra of Pin(4,4), of Spin(4,4) and of their identity component $\mathrm{Spin}_0(4,4)$ is the same, $spin(4,4)$, of dimension $\dim O(4,4) = 28$.

8.2 The scaled commutators are a basis of spin(4,4)

$$S_{ab} = \tfrac14[T_a,T_b] = \tfrac12T_aT_b \quad (a\neq b), \qquad S_{aa} = 0, \qquad S_{ba} = -S_{ab}$$

Symbols in this equation.

- S_{ab}: The scaled commutator $S_{ab} = \frac14[T_a,T_b]$ ($[X,Y] = XY - YX$), the generators of spin(4,4).
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0,\dots,7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- $[X,Y]$, $\{X,Y\}$: The commutator $XY - YX$ and the anticommutator $XY + YX$ of two matrices.
- a,b,c,d: Frame (orthonormal) indices running over $0,1,\dots,7$; the value k belongs to the author's coordinate x_k (x_0 the hidden space-like direction, x_1,x_2,x_3 ordinary space, x_4 the time, x_5,x_6,x_7 the extra time-like directions). No sum over a repeated frame index unless a sum is written.

(`C22_SAB_half_products` (passed)). The 28 matrices S_{ab} with $a < b$ are linearly independent (`C21_SAB_independent` (passed)) and satisfy, for all indices (`C23_SAB_vector_action` (passed), `C24_spin44_closure` (passed)),

$$[S_{ab}, T_c] = \eta_{bc} T_a - \eta_{ac} T_b,$$
$$[S_{ab}, S_{cd}] = \eta_{bc} S_{ad} - \eta_{ac} S_{bd} - \eta_{bd} S_{ac} + \eta_{ad} S_{bc}$$

Symbols in this equation.

- S_{ab}: The scaled commutator $S_{ab} = \frac{1}{4}[T_a, T_b]$ ($[X, Y] = XY - YX$), the generators of spin(4,4).
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- $[X, Y]$, $\{X, Y\}$: The commutator $XY - YX$ and the anticommutator $XY + YX$ of two matrices.
- η_{ab}: The flat frame metric, $\eta = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ in the order $x_0, \ldots, x_7$ (the author's `eta4488`); η^{ab} is the same matrix.
- a, b, c, d: Frame (orthonormal) indices running over $0, 1, \ldots, 7$; the value k belongs to the author's coordinate x_k (x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the extra time-like directions). No sum over a repeated frame index unless a sum is written.

The first relation follows from the theorem of the previous section in one line: $[T_a T_b, T_c] = T_a\{T_b, T_c\} - \{T_a, T_c\}T_b = 2\eta_{bc} T_a - 2\eta_{ac} T_b$. It says that $\mathrm{ad}(S_{ab})$ acts on the span of the T_c by the matrix

$$(L_{ab})^c{}_d = \delta^c_a \, \eta_{bd} - \delta^c_b \, \eta_{ad}, \qquad L_{ab}^{\mathsf{T}} \eta + \eta L_{ab} = 0$$

Symbols in this equation.

- L_{ab}: The generator of $O(4,4)$ on vectors in the plane (a, b): $(L_{ab})^c{}_d = \delta^c_a \eta_{bd} - \delta^c_b \eta_{ad}$.
- δ_{pq}: The Kronecker delta: 1 if $p = q$, else 0 (the author's `ID4[[p,q]]`).
- η_{ab}: The flat frame metric, $\eta = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ in the order $x_0, \ldots, x_7$ (the author's `eta4488`); η^{ab} is the same matrix.
- a, b, c, d: Frame (orthonormal) indices running over $0, 1, \ldots, 7$; the value k belongs to the author's coordinate x_k (x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the extra time-like directions). No sum over a repeated frame index unless a sum is written.

the generator of $O(4,4)$ in the plane (a, b) (`C25_isomorphism_so44` (passed)). The second relation, with the structure constants of $so(4,4)$, shows that $S_{ab} \mapsto L_{ab}$ is an isomorphism of Lie algebras $\mathrm{span}\{S_{ab}\} \cong so(4,4)$. Since every $\exp(\theta S_{ab})$ lies in Pin(4,4) (next paragraph) and $\frac{\mathrm{d}}{\mathrm{d}\theta} \exp(\theta S_{ab})|_0 = S_{ab}$, the 28 independent S_{ab} lie in the 28-dimensional Lie algebra of Pin(4,4) and are a basis of it: $spin(4,4) = \mathrm{span}_{\mathbb{R}}\{S_{ab}\}$.

8.3 The exponentials are products of two unit vectors

$(T_a T_b)^2 = -T_a^2 T_b^2 = -\eta_{aa}\eta_{bb} I_{16}$ for $a \neq b$, so the exponential series sums to (`C26_exponentials_two_unit_vectors` (passed))

$$\exp(\theta S_{ab}) = \begin{cases} \cos\frac{\theta}{2}\, I_{16} + \sin\frac{\theta}{2}\, T_a T_b, & \eta_{aa} = \eta_{bb} \text{ (a rotation)}, \\ \cosh\frac{\theta}{2}\, I_{16} + \sinh\frac{\theta}{2}\, T_a T_b, & \eta_{aa} = -\eta_{bb} \text{ (a boost)} \end{cases}$$

Symbols in this equation.

- S_{ab}: The scaled commutator $S_{ab} = \frac{1}{4}[T_a, T_b]$ ($[X, Y] = XY - YX$), the generators of spin(4,4).
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- θ: A real parameter (an angle for a rotation, a rapidity for a boost).
- η_{ab}: The flat frame metric, $\eta = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ in the order $x_0, \ldots, x_7$ (the author's `eta4488`); η^{ab} is the same matrix.
- I_n: The $n \times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).

and each is a product $u_1 u_2$ of two unit vectors: for $\eta_{aa} = +1$, $u_1 = T_a$ and $u_2 = \cos\frac{\theta}{2} T_a + \sin\frac{\theta}{2} T_b$ (rotation) or $u_2 = \cosh\frac{\theta}{2} T_a + \sinh\frac{\theta}{2} T_b$ (boost); for $\eta_{aa} = -1$, $u_1 = -T_a$ and the sign of the T_b term reversed. Hence $\exp(\theta S_{ab}) \in \mathrm{Spin}(4,4) \subset \mathrm{Pin}(4,4)$.

8.4 What the exponentials generate

Theorem. The group generated by $\{\exp(\theta S_{ab}) : \theta \in \mathbb{R},\ 0 \le a < b \le 7\}$ is the identity component $\mathrm{Spin}_0(4,4)$ of Pin(4,4).

Proof. (i) Every $\exp(\theta S_{ab})$ is joined to I_{16} by the path $\theta' \mapsto \exp(\theta' S_{ab})$, $0 \le \theta' \le \theta$, inside Pin(4,4), so it and every product of such elements lie in the identity component; their images $\rho = \exp(\theta L_{ab})$ in $O(4,4)$ have the determinants of both diagonal blocks (the space-like block $x_0..x_3$ and the time-like block $x_4..x_7$) equal to $+1$ for rotations and $\cosh\theta > 0$ for boosts, i.e. they lie in the identity component $SO_0(4,4)$ (`C28_exponentials_in_identity_component` (passed)). (ii) The map $(\theta_1, \ldots, \theta_{28}) \mapsto \prod_k \exp(\theta_k S_k)$, S_k running over the 28 basis elements, has at 0 the differential $(\theta_k) \mapsto \sum_k \theta_k S_k$, an isomorphism onto the Lie algebra; by the inverse function theorem its image contains a neighbourhood of I_{16} in $\mathrm{Spin}_0(4,4)$. A connected group is generated by any neighbourhood of its identity, so the generated group is all of $\mathrm{Spin}_0(4,4)$. □

The four components of Pin(4,4). The twisted adjoint images of I_{16}, T_0, T_4 and T_0T_4 in $O(4,4)$ have the following signs of the determinants of the space-like block ($x_0..x_3$) and of the time-like block ($x_4..x_7$) (`C27_pin44_components` (passed)):

element g	sign det (space-like block)	sign det (time-like block)	det $\rho(g)$	component
I_{16}	+1	+1	+1	$\mathrm{Spin}_0(4,4)$
T_0	-1	+1	-1	$\mathrm{Spin}_0(4,4)\,T_0$
T_4	+1	-1	-1	$\mathrm{Spin}_0(4,4)\,T_4$
T_0T_4	-1	-1	+1	$\mathrm{Spin}_0(4,4)\,T_0T_4$

The four sign pairs are different, and the block signs are constant on connected pieces of $O(4,4)$, so the four elements lie in the four different components of Pin(4,4), each of which is a coset of the identity component:

$$\mathrm{Pin}(4,4) = \mathrm{Spin}_0(4,4) \ \cup\ \mathrm{Spin}_0(4,4)\,T_0 \ \cup\ \mathrm{Spin}_0(4,4)\,T_4 \ \cup\ \mathrm{Spin}_0(4,4)\,T_0T_4 = \big\langle \exp(\theta S_{ab}),\ T_0,\ T_4 \big\rangle$$

Symbols in this equation.

- S_{ab}: The scaled commutator $S_{ab} = \frac{1}{4}[T_a, T_b]$ ($[X,Y] = XY - YX$), the generators of spin(4,4).
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- θ: A real parameter (an angle for a rotation, a rapidity for a boost).

($\langle \cdots \rangle$ is the group generated). Spin(4,4) is the union of the first and the last coset. So the scaled commutators generate Pin(4,4) as its Lie algebra and, through their exponentials, its identity component; the reflections T_0 (space-like) and T_4 (time-like) supply the other three components, which no product of exponentials can reach.

Irreducibility. The only matrices commuting with all eight T_a are the multiples of I_{16} (`C18_pin44_irreducible` (passed)): $\mathbb{C}^{16}$ is an irreducible representation of Pin(4,4). The matrices commuting with all S_{ab} form the two-dimensional algebra spanned by I_{16} and Γ (`C17_commutant_spin44` (passed)); under Spin(4,4) the space splits into the two halves $P_L\mathbb{C}^{16}$ and $P_R\mathbb{C}^{16}$, each irreducible, and the two are inequivalent (no non-zero intertwiner; `C19_halves_irreducible_inequivalent` (passed)).

9 Chirality: re-derivation of the projection operators

The author's subsection "Left - Handed vs . Right - Handed Fermion Fields (Chirality) ; projection operators" (cells 160 to 166) reads, verbatim:

```
cell 160  Subsection
    Left - Handed vs . Right - Handed Fermion Fields (Chirality) ; projection operators
cell 161  Input In[111]:=
    (Subscript[P, L]=1/2(ID16-T16^A[8]))//MatrixForm
cell 162  Output Out[111]//MatrixForm=
```

```
    (\[NoBreak]{{1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    ↪  0, 0, 0}, {0, 0, 1 ...
cell 163  Input In[112]:=
    (Subscript[P, R]=1/2(ID16+T16^A[8]))//MatrixForm
cell 164  Output Out[112]//MatrixForm=
    (\[NoBreak]{{0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
    ↪  0, 0, 0}, {0, 0, 0 ...
cell 165  Input In[113]:=
    {ID16== Subscript[P, L]+Subscript[P, R],Subscript[P, L].Subscript[P, L]==Subscript[P, L],Subscript[P,
    ↪  L].Subscript[P, R]==Subscript[P, R].Subscript[P, L]==ZERO16}
cell 166  Output Out[113]=
    {True,True,True} ...
```

Step 1: the chirality matrix. A chirality operator must commute with every Lorentz generator S_{ab} (so that it is the same in every frame), square to the identity, and not be a multiple of I_{16}. The commutant of the S_{ab} is $\{xI_{16} + y\Gamma\}$ with (`C17_commutant_spin44` (passed), `C10_chirality_matrix` (passed), `C11_chirality_commutes_with_SAB` (passed))

$$\Gamma = T_0T_1T_2T_3T_4T_5T_6T_7, \qquad \Gamma^2 = (-1)^{8\cdot 7/2}\,\eta_{00}\eta_{11}\cdots\eta_{77}\,I_{16} = I_{16}, \qquad \Gamma T_a = -T_a\Gamma$$

Symbols in this equation.

- Γ: The chirality matrix, the author's `T16^A[8]` $= T_0T_1\cdots T_7$.
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0,\ldots,7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- η_{ab}: The flat frame metric, $\eta = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ in the order $x_0,\ldots,x_7$ (the author's `eta4488`); η^{ab} is the same matrix.
- I_n: The $n\times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).

(moving the eight factors of $\Gamma\Gamma$ past each other gives the sign $(-1)^{28} = +1$, and the product of the squares is $(+1)^4(-1)^4 = +1$; `C12_chirality_square_formula` (passed)). Γ anticommutes with every T_a because T_a anticommutes with the seven other factors and commutes with itself (seven sign changes). Explicitly (`B8_stored_T16A8` (passed))

$$\Gamma = \begin{pmatrix} -I_8 & 0 \\ 0 & I_8 \end{pmatrix} = \begin{pmatrix}
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

Symbols in this equation.

- Γ: The chirality matrix, the author's `T16^A[8]` $= T_0T_1\cdots T_7$.
- I_n: The $n\times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).
- 0: The zero matrix of the size required by its position (in a block matrix: a zero block).

Step 2: the projectors. A projector $X = xI_{16} + y\Gamma$ satisfies $X^2 = X$, i.e. $x^2 + y^2 = x$ and $2xy = y$; the solutions are $(x, y) = (0, 0)$, $(1, 0)$, $(\frac{1}{2}, -\frac{1}{2})$ and $(\frac{1}{2}, \frac{1}{2})$ (`C20_idempotents_of_chirality_algebra` (passed)). The two non-trivial ones are the author's

$$P_L = \tfrac{1}{2}\big(I_{16} - \Gamma\big) = \begin{pmatrix} I_8 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_R = \tfrac{1}{2}\big(I_{16} + \Gamma\big) = \begin{pmatrix} 0 & 0 \\ 0 & I_8 \end{pmatrix}$$

Symbols in this equation.

- $P_L,\ P_R$: The author's chiral projection operators `P_L` $= \frac{1}{2}(I_{16} - \Gamma)$ and `P_R` $= \frac{1}{2}(I_{16} + \Gamma)$ (left-handed and right-handed).
- Γ: The chirality matrix, the author's `T16^A[8]` $= T_0 T_1 \cdots T_7$.
- I_n: The $n \times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).
- 0: The zero matrix of the size required by its position (in a block matrix: a zero block).

equal, entry by entry, to the matrices the notebook stored (`B9_stored_PL` (passed), `B10_stored_PR` (passed)):

$$P_L = \begin{pmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

$$P_R = \begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

Symbols in this equation.

- $P_L,\ P_R$: The author's chiral projection operators `P_L` $= \frac{1}{2}(I_{16} - \Gamma)$ and `P_R` $= \frac{1}{2}(I_{16} + \Gamma)$ (left-handed and right-handed).

Step 3: their properties (`C13_projectors` (passed), `C14_projectors_explicit` (passed), `C15_projectors_and_gammas` (passed), `C16_projectors_and_adjoint` (passed)): the author's three tests of cell 165 and more,

$$P_L + P_R = I_{16}, \quad P_L^2 = P_L, \quad P_R^2 = P_R, \quad P_L P_R = P_R P_L = 0, \quad \mathrm{Tr}\, P_L = \mathrm{Tr}\, P_R = 8,$$
$$P_L T_a = T_a P_R, \qquad [P_L, S_{ab}] = 0, \qquad [P_L, \sigma_{16}] = 0, \qquad P_L^{\mathsf{T}} = P_L$$

Symbols in this equation.

- $P_L,\ P_R$: The author's chiral projection operators `P_L` $= \frac{1}{2}(I_{16} - \Gamma)$ and `P_R` $= \frac{1}{2}(I_{16} + \Gamma)$ (left-handed and right-handed).
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- S_{ab}: The scaled commutator $S_{ab} = \frac{1}{4}[T_a, T_b]$ ($[X, Y] = XY - YX$), the generators of spin(4,4).
- σ_{16}: The author's `sigma16` $= T_0 T_1 T_2 T_3$, the matrix of the Dirac adjoint $\bar{\Psi} = \Psi^\dagger \sigma_{16}$.
- $[X, Y],\ \{X, Y\}$: The commutator $XY - YX$ and the anticommutator $XY + YX$ of two matrices.
- $X^{\mathsf{T}},\ X^\dagger$: The transpose of a matrix and its Hermitian conjugate (transpose of the complex conjugate); for the real matrices of this document they coincide.
- I_n: The $n \times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).

So the left-handed part $P_L\Psi$ consists of the components $\Psi_0, \ldots, \Psi_7$ and the right-handed part $P_R\Psi$ of $\Psi_8, \ldots, \Psi_{15}$; each is an irreducible representation of Spin(4,4), the two inequivalent; every Dirac matrix exchanges them. Two consequences matter for every Lagrangian built with these matrices: because σ_{16} commutes with P_L,

$$\overline{P_L \Psi} = \bar{\Psi} P_L, \qquad \bar{\Psi}\Psi = \bar{\Psi} P_L \Psi + \bar{\Psi} P_R \Psi, \qquad \bar{\Psi} T_a \Psi = \bar{\Psi} P_L T_a P_R \Psi + \bar{\Psi} P_R T_a P_L \Psi$$

Symbols in this equation.

- $\bar{\Psi}$: The Dirac adjoint $\bar{\Psi} = \Psi^\dagger \sigma_{16}$.
- Ψ: The 16-component dirac16complex spinor field (components $\Psi_0, \ldots, \Psi_{15}$), on which the matrices act.
- $P_L,\ P_R$: The author's chiral projection operators `P_L` $= \frac{1}{2}(I_{16} - \Gamma)$ and `P_R` $= \frac{1}{2}(I_{16} + \Gamma)$ (left-handed and right-handed).
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.

a bilinear with an even number of Dirac matrices (such as the mass term $\bar{\Psi}\Psi$) never mixes the two chiralities, while one with an odd number (such as the kinetic term $\bar{\Psi} T_a \partial \Psi$) always connects the left-handed with the right-handed part. In this signature the Dirac adjoint is built with the even matrix σ_{16}; in four-dimensional Minkowski space it is built with the odd γ^0, which is why there the mass term mixes the chiralities and the kinetic term does not.

10 Other important 16×16 matrices of the notebook

The complex structure $J = T_4$ (the author's cells 654 to 669): $J^2 = -I_{16}$ and $J^{\mathsf{T}} \sigma_{16} J = \sigma_{16}$, the author's `In[476]` (`C29_complex_structure_T4` (passed)); the author notes that he uses x_4 (not the periodic x_0) for it.

$$J = T_4, \qquad J^2 = -I_{16}, \qquad J^{\mathsf{T}} \sigma_{16} J = \sigma_{16}$$

Symbols in this equation.

- J: The author's complex structure $J = T_4$ (the Dirac matrix of the time x_4), $J^2 = -I_{16}$.
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` ($a = 0, \ldots, 7$), the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.
- σ_{16}: The author's `sigma16` $= T_0 T_1 T_2 T_3$, the matrix of the Dirac adjoint $\bar{\Psi} = \Psi^\dagger \sigma_{16}$.

- X^{T}, $X^{\dagger}$: The transpose of a matrix and its Hermitian conjugate (transpose of the complex conjugate); for the real matrices of this document they coincide.
- I_n: The $n \times n$ identity matrix ($n = 4, 8, 16$; the author's `ID4`, `ID8`, `ID16`).

The product $T_1T_2T_3T_4T_5T_6$ stored by the author (`Out[489]`) is reproduced (`B12_stored_T1_to_T6` (passed)); the products $T_4T_5T_6T_7T_a$ are symmetric (`Out[100]`) and $\Gamma = \sigma_{16}T_4T_5T_6T_7$ (`Out[101]`, `Out[498]`), all reproduced (`B13_stored_boolean_outputs` (passed)).

The basis of all real 16×16 **matrices** is the set of the 256 ordered products of the T_a, with $\mathrm{Tr}(MM)/16 = \pm 1$ for each (the author's `Out[202]`) and 136 symmetric, 120 antisymmetric members (`C30_basis_256` (passed)).

11 Agreement with every stored output of the matrix cells

The notebook file keeps the outputs that Mathematica computed when the author evaluated the cells. Each was parsed and compared with the rebuilt matrices (matrices entry by entry; True/False lists value by value):

cell	label	what	agrees
12	`Out[9]//MatrixForm=`	eta4488	yes
35	`Out[39]=`	s4by4[h], t4by4[h], h = 1..3	yes
74	`Out[63]//MatrixForm=`	zeta[7]	yes
100	`Out[81]//MatrixForm=`	tau[7]	yes
102	`Out[82]//MatrixForm=`	Omega = sigma.tau[7]	yes
106	`Out[84]//MatrixForm=`	tau[5].tau[6].tau[7]	yes
108	`Out[85]//MatrixForm=`	sigma.Omega	yes
135	`Out[98]//MatrixForm=`	T16^A[8]	yes
162	`Out[111]//MatrixForm=`	P_L	yes
164	`Out[112]//MatrixForm=`	P_R	yes
313	`Out[210]=`	T16^A[0..7]	yes
791	`Out[489]//MatrixForm=`	$\mathrm{T16}^{\mathrm{A[1]}\ldots\mathrm{T16}}$A[6]	yes
76	`Out[64]=`	sigma == zeta[1].zeta[2].zeta[3]	yes
78	`Out[65]=`	1/2{zeta[A], zeta[B]} == -eta[A,B] ID8, A, B = 1..7 (49 cases)	yes
82	`Out[67]=`	-eta[A,A] zeta[A] == Transpose[zeta[A]]	yes
89	`Out[72]//MatrixForm=`	sigma == tau[1].tau[2].tau[3]	yes
93	`Out[76]=`	sigma.taubar[a] == Transpose[sigma.tau[a]]	yes
95	`Out[77]//MatrixForm=`	1/2(tau[A].taubar[B] + tau[B].taubar[A]) == eta[A,B] ID8 (64 cases)	yes
104	`Out[83]=`	Omega == tau[4].tau[5].tau[6]	yes
112	`Out[87]=`	tau[1]...tau[7] == tau[0] == ID8	yes
116	`Out[89]=`	sigma == tau[4].tau[5].tau[6].tau[7]	yes
120	`Out[90]=`	tau[1].tau[2].tau[3].tau[a] antisymmetric	yes

cell	label	what	agrees
124	`Out[92]=`	tau[5].tau[6].tau[7].taubar[a] == Transpose[tau[5].tau[6].tau[7].tau[a]]	yes
126	`Out[93]=`	tau[4].tau[a] symmetric	yes
128	`Out[94]=`	tau[4].taubar[a] == Transpose[tau[4].tau[a]]	yes
137	`Out[99]=`	T0.T1.T2.T3.T[a] antisymmetric	yes
139	`Out[100]=`	T4.T5.T6.T7.T[a] symmetric, a = 0..8	yes
141	`Out[101]=`	T16^A[8] == sigma16.T4.T5.T6.T7	yes
143	`Out[102]=`	T16^A[a] symmetric, a = 0..8	yes
145	`Out[103]=`	T16^A[a] antisymmetric, a = 0..8	yes
147	`Out[104]=`	sigma16.T16^A[a] antisymmetric	yes
149	`Out[105]=`	sigma16 == ArrayFlatten[{{-sigma,0},{0,sigma}}]	yes
157	`Out[109]//MatrixForm=`	1/2{T16^A[A], T16^A[B]} == eta4488[[A+1,B+1]] ID16 (64 cases)	yes
159	`Out[498]=`	T16^A[8] === sigma16.T4.T5.T6.T7	yes
166	`Out[113]=`	{ID16 == P_L + P_R, P_L.P_L == P_L, P_L.P_R == P_R.P_L == ZERO16}	yes
280	`Out[188]=`	Id16 == $\mathrm{T16}^{\mathrm{A[0]}\ldots\mathrm{T16}}$A[8]	yes
655	`Out[392]=`	$\mathrm{T16}^{\mathrm{A[4].T16}}$A[4] === -Id16	yes
775	`Out[476]=`	J = T16^A[4]: {J.J === -ID16, sigma16 === J^T.sigma16.J, !(sigma16.J antisymmetric)}	yes

12 The same matrices in every calculation

The author asked to verify that the Dirac matrices used in the calculations are those of `4+4.nb`. Every other source of Dirac matrices in this repository was compared with them, entry by entry:

source	equal to the T_a of 4+4.nb
provenance/dirac_matrices/author_notebook_T16.json (the Pair_Creation notebook, evaluated by WolframScript 15.0.1)	yes, all eight
Revision/algebra/gammas.json (Revision record)	yes, all eight
universe_model_2/outputs/algebra.json (UM-2)	yes, all eight

source	equal to the T_a of 4+4.nb
Revision/field_equations_a4/wolfram/FieldEquationsA4.wl (Wolfram package, evaluated by WolframScript)	yes, all eight
Revision/algebra/wolfram/RevisionAlgebra.wl (Wolfram package, evaluated by WolframScript)	yes, all eight
wolfram/Dirac16ComplexAlgebra.wl (Wolfram package, evaluated by WolframScript)	yes, all eight
wolfram/Dirac16ComplexGeometry.wl (Wolfram package, evaluated by WolframScript)	yes, all eight
wolfram/Dirac16ComplexPrimordial.wl (Wolfram package, evaluated by WolframScript)	yes, all eight
wolfram/Dirac16ComplexKohnSham.wl (Wolfram package, evaluated by WolframScript)	yes, all eight
wolfram/Dirac16ComplexPairing.wl (Wolfram package, evaluated by WolframScript)	yes, all eight
wolfram/Dirac16ComplexMatterAntimatter.wl (Wolfram package, evaluated by WolframScript)	yes, all eight
wolfram/Dirac16Complex00.wl (Wolfram package, evaluated by WolframScript)	yes, all eight
artifacts/dirac16complex/arbitrary-field/algebra-fixture.json (Stage 1 fixture)	yes, all eight

The author's notebook `Pair_Creation_of_Universes_WaveFunctionOfUniverse-4+4-Einstein-Lovelock-Nash.nb` builds the matrices with the same formulas (there τ is written directly with the formulas that `4+4.nb` gives to its intermediate ζ; `D2_pair_creation_same_formulas` (passed)), and its evaluation by WolframScript in an earlier session gave $T_0, \ldots, T_7$, Γ, σ_{16}, P_L, P_R and all S_{ab} equal to those of `4+4.nb` (`D1_pair_creation_extraction` (passed)). In UM-3 every computation imports its matrices from `universe_model_3/code/um3_gammas.py`, the cell-by-cell transcription checked here, so UM-3 uses exactly these matrices.

13 How to reproduce

Requirements: Python 3 with sympy (`python -m pip install sympy`); for the PDF, pandoc and a TeX distribution. From the repository root:

```
python universe_model_3/code/um3_dirac_matrices.py       # 55 checks; writes outputs/dirac_matrices.json
↪  and reports/dirac_matrices_checks.json
python universe_model_3/code/um3_dirac_matrices_doc.py   # writes "universe_model_3/dirac matrices.md"
python universe_model_3/tools/build_docs.py "dirac matrices"   # writes the .tex and the .pdf
```

The first program needs no Mathematica: it reads the notebook file with `universe_model_3/tools/nb_reader.py`. Its run time was 36 s.

14 Appendix: the 28 generators

Each $2S_{ab} = T_aT_b$ $(a < b)$ is a signed permutation matrix; it is listed by its non-zero entries "row:sign column". Rotations ($\eta_{aa} = \eta_{bb}$, 12 of them) satisfy $(T_aT_b)^2 = -I_{16}$, boosts ($\eta_{aa} = -\eta_{bb}$, 16) $(T_aT_b)^2 = +I_{16}$.

$$2S_{ab} = T_aT_b \qquad (0 \le a < b \le 7)$$

Symbols in this equation.

- S_{ab}: The scaled commutator $S_{ab} = \frac{1}{4}[T_a, T_b]$ ($[X, Y] = XY - YX$), the generators of spin(4,4).
- T_a: The author's eight real 16×16 Dirac matrices `T16^A[a]` $(a = 0, \ldots, 7)$, the gamma matrix of the frame direction of x_a: $\gamma^{(x_a)} = T_a$.

- a, b, c, d: Frame (orthonormal) indices running over $0, 1, \ldots, 7$; the value k belongs to the author's coordinate x_k (x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the extra time-like directions). No sum over a repeated frame index unless a sum is written.
- $2S_{01} = T_0T_1$ (rotation): 0:+7, 1:+6, 2:-5, 3:-4, 4:+3, 5:+2, 6:-1, 7:-0, 8:-15, 9:-14, 10:+13, 11:+12, 12:-11, 13:-10, 14:+9, 15:+8.
- $2S_{02} = T_0T_2$ (rotation): 0:-6, 1:+7, 2:+4, 3:-5, 4:-2, 5:+3, 6:+0, 7:-1, 8:+14, 9:-15, 10:-12, 11:+13, 12:+10, 13:-11, 14:-8, 15:+9.
- $2S_{03} = T_0T_3$ (rotation): 0:+5, 1:-4, 2:+7, 3:-6, 4:+1, 5:-0, 6:+3, 7:-2, 8:-13, 9:+12, 10:-15, 11:+14, 12:-9, 13:+8, 14:-11, 15:+10.
- $2S_{04} = T_0T_4$ (boost): 0:+5, 1:-4, 2:-7, 3:+6, 4:-1, 5:+0, 6:+3, 7:-2, 8:-13, 9:+12, 10:+15, 11:-14, 12:+9, 13:-8, 14:-11, 15:+10.
- $2S_{05} = T_0T_5$ (boost): 0:-6, 1:-7, 2:+4, 3:+5, 4:+2, 5:+3, 6:-0, 7:-1, 8:+14, 9:+15, 10:-12, 11:-13, 12:-10, 13:-11, 14:+8, 15:+9.
- $2S_{06} = T_0T_6$ (boost): 0:-7, 1:+6, 2:-5, 3:+4, 4:+3, 5:-2, 6:+1, 7:-0, 8:+15, 9:-14, 10:+13, 11:-12, 12:-11, 13:+10, 14:-9, 15:+8.
- $2S_{07} = T_0T_7$ (boost): 0:-0, 1:-1, 2:-2, 3:-3, 4:+4, 5:+5, 6:+6, 7:+7, 8:+8, 9:+9, 10:+10, 11:+11, 12:-12, 13:-13, 14:-14, 15:-15.
- $2S_{12} = T_1T_2$ (rotation): 0:+1, 1:-0, 2:+3, 3:-2, 4:+5, 5:-4, 6:+7, 7:-6, 8:+9, 9:-8, 10:+11, 11:-10, 12:+13, 13:-12, 14:+15, 15:-14.
- $2S_{13} = T_1T_3$ (rotation): 0:+2, 1:-3, 2:-0, 3:+1, 4:+6, 5:-7, 6:-4, 7:+5, 8:+10, 9:-11, 10:-8, 11:+9, 12:+14, 13:-15, 14:-12, 15:+13.
- $2S_{14} = T_1T_4$ (boost): 0:+2, 1:-3, 2:+0, 3:-1, 4:-6, 5:+7, 6:-4, 7:+5, 8:+10, 9:-11, 10:+8, 11:-9, 12:-14, 13:+15, 14:-12, 15:+13.
- $2S_{15} = T_1T_5$ (boost): 0:+1, 1:+0, 2:+3, 3:+2, 4:-5, 5:-4, 6:-7, 7:-6, 8:+9, 9:+8, 10:+11, 11:+10, 12:-13, 13:-12, 14:-15, 15:-14.
- $2S_{16} = T_1T_6$ (boost): 0:+0, 1:-1, 2:-2, 3:+3, 4:-4, 5:+5, 6:+6, 7:-7, 8:+8, 9:-9, 10:-10, 11:+11, 12:-12, 13:+13, 14:+14, 15:-15.
- $2S_{17} = T_1T_7$ (boost): 0:-7, 1:-6, 2:+5, 3:+4, 4:+3, 5:+2, 6:-1, 7:-0, 8:-15, 9:-14, 10:+13, 11:+12, 12:+11, 13:+10, 14:-9, 15:-8.
- $2S_{23} = T_2T_3$ (rotation): 0:+3, 1:+2, 2:-1, 3:-0, 4:+7, 5:+6, 6:-5, 7:-4, 8:+11, 9:+10, 10:-9, 11:-8, 12:+15, 13:+14, 14:-13, 15:-12.
- $2S_{24} = T_2T_4$ (boost): 0:+3, 1:+2, 2:+1, 3:+0, 4:-7, 5:-6, 6:-5, 7:-4, 8:+11, 9:+10, 10:+9, 11:+8, 12:-15, 13:-14, 14:-13, 15:-12.
- $2S_{25} = T_2T_5$ (boost): 0:-0, 1:+1, 2:-2, 3:+3, 4:+4, 5:-5, 6:+6, 7:-7, 8:-8, 9:+9, 10:-10, 11:+11, 12:+12, 13:-13, 14:+14, 15:-15.
- $2S_{26} = T_2T_6$ (boost): 0:+1, 1:+0, 2:-3, 3:-2, 4:-5, 5:-4, 6:+7, 7:+6, 8:+9, 9:+8, 10:-11, 11:-10, 12:-13, 13:-12, 14:+15, 15:+14.
- $2S_{27} = T_2T_7$ (boost): 0:+6, 1:-7, 2:-4, 3:+5, 4:-2, 5:+3, 6:+0, 7:-1, 8:+14, 9:-15, 10:-12, 11:+13, 12:-10, 13:+11, 14:+8, 15:-9.
- $2S_{34} = T_3T_4$ (boost): 0:-0, 1:-1, 2:+2, 3:+3, 4:+4, 5:+5, 6:-6, 7:-7, 8:-8, 9:-9, 10:+10, 11:+11, 12:+12, 13:+13, 14:-14, 15:-15.
- $2S_{35} = T_3T_5$ (boost): 0:-3, 1:+2, 2:+1, 3:-0, 4:+7, 5:-6, 6:-5, 7:+4, 8:-11, 9:+10, 10:+9, 11:-8, 12:+15, 13:-14, 14:-13, 15:+12.
- $2S_{36} = T_3T_6$ (boost): 0:+2, 1:+3, 2:+0, 3:+1, 4:-6, 5:-7, 6:-4, 7:-5, 8:+10, 9:+11, 10:+8, 11:+9, 12:-14, 13:-15, 14:-12, 15:-13.

- $2S_{37} = T_3T_7$ (boost): 0:-5, 1:+4, 2:-7, 3:+6, 4:+1, 5:-0, 6:+3, 7:-2, 8:-13, 9:+12, 10:-15, 11:+14, 12:+9, 13:-8, 14:+11, 15:-10.
- $2S_{45} = T_4T_5$ (rotation): 0:-3, 1:+2, 2:-1, 3:+0, 4:-7, 5:+6, 6:-5, 7:+4, 8:-11, 9:+10, 10:-9, 11:+8, 12:-15, 13:+14, 14:-13, 15:+12.
- $2S_{46} = T_4T_6$ (rotation): 0:+2, 1:+3, 2:-0, 3:-1, 4:+6, 5:+7, 6:-4, 7:-5, 8:+10, 9:+11, 10:-8, 11:-9, 12:+14, 13:+15, 14:-12, 15:-13.
- $2S_{47} = T_4T_7$ (rotation): 0:-5, 1:+4, 2:+7, 3:-6, 4:-1, 5:+0, 6:+3, 7:-2, 8:-13, 9:+12, 10:+15, 11:-14, 12:-9, 13:+8, 14:+11, 15:-10.
- $2S_{56} = T_5T_6$ (rotation): 0:+1, 1:-0, 2:-3, 3:+2, 4:+5, 5:-4, 6:-7, 7:+6, 8:+9, 9:-8, 10:-11, 11:+10, 12:+13, 13:-12, 14:-15, 15:+14.
- $2S_{57} = T_5T_7$ (rotation): 0:+6, 1:+7, 2:-4, 3:-5, 4:+2, 5:+3, 6:-0, 7:-1, 8:+14, 9:+15, 10:-12, 11:-13, 12:+10, 13:+11, 14:-8, 15:-9.
- $2S_{67} = T_6T_7$ (rotation): 0:+7, 1:-6, 2:+5, 3:-4, 4:+3, 5:-2, 6:+1, 7:-0, 8:+15, 9:-14, 10:+13, 11:-12, 12:+11, 13:-10, 14:+9, 15:-8.